MIAE 221 · Materials Science & Engineering

Midterm 1-Page Document Cheatsheet & Exam Survival Guide

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Curriculum Grounded in Callister (Chapters 1–7.4) & Lecture Modules 1–12 (Drs. Mamoun Medraj & Martin Pugh)

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1. Bonding, Potential Energy Wells & Physical Property Correlations

- **Net Force & Equilibrium Spacing (r_0)**: $F_N(r) = F_A(r) + F_R(r) = 0$ at $r = r_0$. Net interatomic potential energy: $E_N(r) = \int F_N dr = E_A(r) + E_R(r)$.
- **Potential Energy Well Depth (E_0)**:
  - **Melting Temperature (T_m)**: Deeper energy trough $E_0 \implies$ higher bond energy $\implies$ higher T_m ($E_0 \propto T_m$).
  - **Modulus of Elasticity (E)**: Curvature at trough minimum: $E \propto \left(\frac{d^2 E_N}{dr^2}\right)_{r_0} = \left(\frac{d F_N}{dr}\right)_{r_0}$. $Steeper slope \implies$ stiffer material.
  - **Thermal Expansion (α_L)**: Governed by the **asymmetry** of the potential energy well. Highly symmetric well $\implies$ lower α_L. Ceramics (deep, symmetric) have much lower α_L than polymers.
- **Primary Bonding Types**:
  - **Ionic**: Electron transfer between electropositive metal and electronegative nonmetal; nondirectional; high E_0 and T_m; brittle; electrical insulator.
  - **Covalent**: Electron sharing between atoms with small electronegativity differences; **directional** (fixed bond angles); diamond, polymers (chain backbone).
  - **Metallic**: Positive ion cores surrounded by a delocalized "sea of electrons"; nondirectional; good electrical/thermal conductors; ductile (slip without bond fracture).
- **Secondary (van der Waals & Hydrogen) Bonding**:
  - Dipole interactions (permanent or fluctuating). **Hydrogen bonding** (H bonded to F, O, or N) is the strongest secondary bond ($E \sim 50\ \mathrm{kJ/mol}$). Holds adjacent polymer chains together.

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2. Crystal Lattices, Unit Cell Geometry & Density

Crystal Structure	Coordinate Number (CN)	Atom Count (n)	Lattice Parameter $a(R)$	Atomic Packing Factor (APF)	Close-Packed Direction	Close-Packed Plane
FCC (Cu, Al, Ni, Au)	12	4	$a = 2R\sqrt{2}$	0.74	$\langle 110 \rangle$	111
BCC (α-Fe, Cr, Mo, W)	8	2	$a = \frac{4R}{\sqrt{3}}$	0.68	$\langle 111 \rangle$	110 (densest)
HCP (Ti, Mg, Zn, Co)	12	6	$a = 2R,; c/a = 1.633$	0.74	$\langle 11\bar{2}0 \rangle$	(0001) (basal)

- **Theoretical Density Formula**:

$$\rho = \frac{n \cdot A}{V_c \cdot N_A} \quad \left[\mathrm{g/cm^3}\right]$$

  Where n = atoms/unit cell, A = atomic weight ($\mathrm{g/mol}$), V_c = unit cell volume (a^3 for cubic), and $N_A = 6.022 \times 10^{23}\ \mathrm{atoms/mol}$.
- **Stacking Sequences**:
  - **FCC**: $ABCABC\ldots$ along $[111]$ direction.
  - **HCP**: $ABABAB\ldots$ along $[0001]$ direction.

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3. Crystallographic Directions, Planes & Miller Indices

- **Directions $[uvw]$ (Square Brackets)**:
  1. Position vector tail at coordinate origin $(0, 0, 0)$ (shift origin if vector has negative components).
  2. Read projection coordinates of vector head (x_1, y_1, z_1) in terms of unit cell dimensions a, b, c.
  3. Clear fractions to achieve the smallest integer triad: $[uvw]$. Negative numbers carry an overbar: $[\bar{1}20]$.
  4. Family of equivalent directions: $\langle uvw \rangle$.
- **Planes (hkl) (Round Parentheses)**:
  1. **Crucial Rule**: The plane must **NEVER** pass through the chosen origin. If it does, translate the origin to an adjacent unit cell corner.
  2. Determine axial intercepts in units of a, b, c: (A, B, C). (If plane is parallel to an axis, intercept = ∞).
  3. Take the **reciprocals** of intercepts: $(1/A, 1/B, 1/C)$. Reciprocal of ∞ is 0.
  4. Clear fractions to smallest integers: (hkl). Negative intercepts carry an overbar: $(\bar{1}10)$.
  5. Family of equivalent planes: hkl.
- **Hexagonal Miller-Bravais System**:
  - 4 axes: a_1, a_2, a_3 (at 120°) and z.
  - Planar indices $(hkil)$ where index i must always satisfy: $i = -(h + k)$.
  - Direction indices $[uvtw]$ satisfy: $t = -(u + v)$. Conversion from 3-axis $[u'v'w']$: $u = \frac{1}{3}(2u' - v')$, $v = \frac{1}{3}(2v' - u')$, $w = w'$.

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4. Linear & Planar Densities, X-Ray Diffraction (XRD) & Bragg's Law

- **Linear Density (LD)**:

$$LD = \frac{\text{Number of atomic diameters centered along vector line}}{\text{Length of direction vector } L} \quad [\text{atoms/nm}]$$

- **Planar Density (PD)**:

$$PD = \frac{\text{Number of atomic 2D circles centered on plane}}{\text{Area of plane } A} \quad \left[\text{atoms/nm}^2\right]$$

- **Bragg's Law of Diffraction**:

$$n\lambda = 2d_{hkl}\sin\theta$$

  Where n = order of reflection (usually 1), λ = X-ray wavelength, θ = diffraction (Bragg) angle, and d_{hkl} = interplanar spacing.
- **Cubic Interplanar Spacing (d_{hkl})**:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

- **Diffraction Selection Rules (Systematic Absences)**:
  - **BCC**: Diffraction peaks occur **ONLY** when the sum of indices is even: $h + k + l = 2n$ (Allowed: $(110), (200), (211), (220)$; Forbidden: $(100), (111), (210)$).
  - **FCC**: Diffraction peaks occur **ONLY** when h, k, l are all unmixed (either **all odd** or **all even**): (Allowed: $(111), (200), (220), (311)$; Forbidden: $(110), (210), (211)$).

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5. Imperfections in Solids & Hume-Rothery Rules

- **Point Defects**:
  - **Vacancies**: Unoccupied atomic lattice sites. Equilibrium concentration follows Arrhenius temperature dependence:

$$N_v = N\exp\left(-\frac{Q_v}{kT}\right) \quad \text{or} \quad N_v = N\exp\left(-\frac{Q_v}{RT}\right)$$

    Where $N = \frac{\rho N_A}{A}$ is the total number of atomic sites, Q_v is vacancy formation activation energy, and $k = 8.62 \times 10^{-5}\ \text{eV/atom} \cdot \text{K}$.
  - **Self-Interstitials**: Host atom squeezed into a small interstitial void; creates severe localized lattice distortion; concentration is extremely low in metals.
- **Substitutional vs. Interstitial Solid Solutions**:
  - **Hume-Rothery Rules for Complete Solid Solubility (e.g., Cu-Ni system)**:
    1. **Atomic Size Factor**: Radius difference $|\Delta r| = \frac{|r_{\text{solute}} - r_{\text{solvent}}|}{r_{\text{solvent}}} \times 100$.
    2. **Crystal Structure**: Must possess identical crystal structure (e.g., Cu is FCC and Ni is FCC).
    3. **Electronegativity Difference**: Electronegativities must be comparable (otherwise an intermetallic compound forms rather than a solid solution).

4. **Valency**: A metal dissolves another metal of higher valency more readily than one of lower valency.

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6. Dislocations, Slip Systems & Strengthening Mechanisms

- **Dislocation Types**:
  - **Edge Dislocation**: Extra half-plane of atoms inserted into the lattice. Burgers vector $\vec{b}$ is strictly **perpendicular** ($\perp$) to the dislocation line.
  - **Screw Dislocation**: Helical/spiral ramp lattice distortion produced by shear stress. Burgers vector $\vec{b}$ is strictly **parallel** ($\parallel$) to the dislocation line.
  - **Mixed Dislocation**: Possesses both edge and screw character. Burgers vector $\vec{b}$ is neither $\perp$ nor $\parallel$ to the dislocation line, but maintains constant magnitude and direction everywhere along its length.
- **Slip Systems**:
  - A slip system consists of a **close-packed plane** and a **close-packed direction** (directions of highest linear density).
  - **FCC**: $111\langle 110\rangle \implies 4 \text{ planes} \times 3 \text{ directions} = \mathbf{12} \text{ slip systems}$. High ductility.
  - **BCC**: $110\langle 111\rangle \implies 6 \text{ planes} \times 2 \text{ directions} = 12 \text{ systems}$ (also 211 and 321, total 48 systems). High strength, moderate ductility.
- **Resolved Shear Stress (Schmid's Law)**:

$$\tau_R = \sigma \cos\phi \cos\lambda$$

  Where σ is applied tensile stress, ϕ is angle between tensile axis and plane normal, and λ is angle between tensile axis and slip direction. Yielding initiates when τ_R reaches the **Critical Resolved Shear Stress** (τ_{CRSS}).
  - Yield occurs at lowest applied stress when $\phi = \lambda = 45^\circ \implies \cos\phi\cos\lambda = 0.5 \implies \sigma_y = 2\tau_{\text{CRSS}}$.
  - If $\phi = 90^\circ$ or $\lambda = 90^\circ$, $\tau_R = 0$ (no slip occurs).
- **Primary Strengthening Mechanisms (All Impede Dislocation Motion)**:
  1. **Grain Boundary Strengthening (Hall-Petch Relation)**:

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

     Smaller average grain diameter $d \implies$ higher grain boundary surface area $\implies$ higher barrier to dislocation pile-ups $\implies$ higher yield strength σ_y.
  2. **Solid Solution Strengthening**: Solute atoms (substitutional or interstitial) generate localized tensile/compressive strain fields that pin and impede dislocations.
  3. **Strain Hardening / Cold Work (CW)**:

     Deformation multiplies dislocation density from $\sim 10^5 \text{ cm}^{-2}$ to $\sim 10^{10} \text{ cm}^{-2}$. Dislocation-dislocation interactions and tangles dramatically increase σ_y and UTS, but **reduce ductility** ().

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7. Diffusion in Solids

- **Diffusion Mechanisms**:

    - **Vacancy Diffusion**: Host or substitutional solute atoms jump into adjacent empty lattice vacancies. Requires both vacancy formation and atomic motion energy.
    - **Interstitial Diffusion**: Small impurity atoms (C, H, O, N) jump into adjacent interstitial void spaces. **Much faster** than vacancy diffusion ($D_{\text{interstitial}} \gg D_{\text{vacancy}}$) because interstitial voids are smaller, more numerous, and have lower activation energies.
    - **Interdiffusion in Couples**: In a copper-nickel diffusion couple pressed together below melting points, Cu atoms diffuse into the Ni matrix AND Ni atoms diffuse into the Cu matrix simultaneously across the concentration gradient.
- **Steady-State Diffusion (Fick's 1st Law)**:

$$J = -D\frac{dC}{dx} \quad \left[\text{kg/m}^2 \cdot \text{s or atoms/cm}^2 \cdot \text{s}\right]$$

  Flux J is constant with time. For a linear concentration profile: $J = -D\frac{C_A - C_B}{x_A - x_B}$.
- **Temperature Dependence of Diffusion Coefficient (Arrhenius Law)**:

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right) \implies \ln D = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T}\right)$$

  Plotting $\ln D$ versus $1/T$ yields a straight line with slope $= -Q_d/R$ and intercept $= \ln D_0$.
- **Non-Steady-State Diffusion (Fick's 2nd Law for Semi-Infinite Solid)**:

$$\frac{C(x,t) - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

  Where C_0 = uniform initial bulk concentration, C_s = constant surface concentration, $C(x,t)$ = concentration at depth x after time t, and $\text{erf}(z)$ is the Gaussian Error Function.
    - Constant composition scaling rule: $\frac{x^2}{Dt} = \text{constant} \implies \frac{x_1^2}{D_1 t_1} = \frac{x_2^2}{D_2 t_2}$.

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8. Mechanical Properties, Tensile Testing & Stress-Strain Behavior

- **Engineering Stress & Strain**:

$$\sigma = \frac{F}{A_0} \quad [\text{MPa}], \qquad \epsilon = \frac{l_i - l_0}{l_0} = \frac{\Delta l}{l_0} \quad [\text{dimensionless}]$$

- **Hooke's Law & Elastic Modulus**:

$$\sigma = E\epsilon \qquad \text{and shear stress: } \tau = G\gamma$$

    - Modulus of Elasticity E depends **only on interatomic bond strength**. Cold working, grain size, and alloying have negligible effect on E! (All carbon and low-alloy steels have $E \approx 207$ GPa).
    - **Poisson's Ratio**: $\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$. For isotropic metals, $E = 2G(1+\nu)$, with $\nu \approx 0.33$.
- **Key Landmarks on the Engineering Stress-Strain Curve**:
    1. **Point B (Proportional Limit)**: Onset of deviation from Hooke's linear elasticity.
    2. **Point C (0.2 Offset Yield Strength σ_y)**: Parallel offset line drawn at $\epsilon = 0.002$ intersects the curve at σ_y. Dictates design stress limits.

3. **Point E (Ultimate Tensile Strength, UTS)**: Maximum engineering stress the material sustains before necking begins.
4. **Uniform vs. Non-Uniform Plastic Deformation**:
    - **Uniform Plastic Deformation**: Occurs across the entire gauge length from yield (σ_y, Point C) up to maximum tensile strength (UTS, Point E).
    - **Non-Uniform Plastic Deformation (Necking)**: Occurs **strictly after Point E** until final fracture at Point F. Localized cross-sectional area decreases rapidly.

- **True Stress & True Strain (Valid Up to the Onset of Necking)**:

$$\sigma_T = \sigma(1+\epsilon), \qquad \epsilon_T = \ln(1+\epsilon)$$

- **Ductility Indices**:

- **Modulus of Resilience (U_r) & Toughness**:
    - Resilience $U_r = \int_0^{\epsilon_y} \sigma d\epsilon \approx \frac{1}{2}\sigma_y \epsilon_y = \frac{\sigma_y^2}{2E}$ (elastic energy absorbed per unit volume).
    - Toughness = total area under the entire stress-strain curve up to fracture (energy absorbed before failure).

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9. Top 8 Midterm Pitfalls & Exam Strategy

1. **Force vs. Energy Curve Equilibrium**: Net force curve crosses zero ($F_N = 0$) at r_0; potential energy curve reaches its minimum trough ($dE/dr = 0$) at r_0. Never confuse zero force with zero energy!
2. **Elastic Modulus Invariance**: Question:

    "How does cold working change the elastic modulus E of steel?"

    Answer: It does **NOT** change E. E is dictated by atomic bond stiffness, not dislocation density.
3. **Diffusion Couple Mechanism**: Copper and nickel undergo **interdiffusion** (both atoms diffuse across the boundary). It is never one-sided!
4. **Miller Indices Negative Origin Trap**: Never pass a plane through the origin. Move the origin to an adjacent corner, read intercepts, and invert.
5. **FCC vs. BCC XRD Peaks**: BCC first peak is always (110) ($h+k+l=2$). FCC first peak is always (111) (unmixed odd).
6. **Necking Onset**: Necking begins at **Point E (UTS)**, NOT at the yield point.
7. **Temperature in Kelvin**: In all Arrhenius equations ($\exp(-Q/RT)$ or $\exp(-Q/kT)$), temperature T must be converted to **Kelvin** ($T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$).
8. **Constant R vs. k**: Use $R = 8.314\ \mathrm{J/mol \cdot K}$ when Q is in $\mathrm{J/mol}$ or $\mathrm{kJ/mol}$; use $k = 8.62 \times 10^{-5}\ \mathrm{eV/atom \cdot K}$ when Q is in $\mathrm{eV/atom}$.